EXP NO 8️⃣
PROGRAM:-

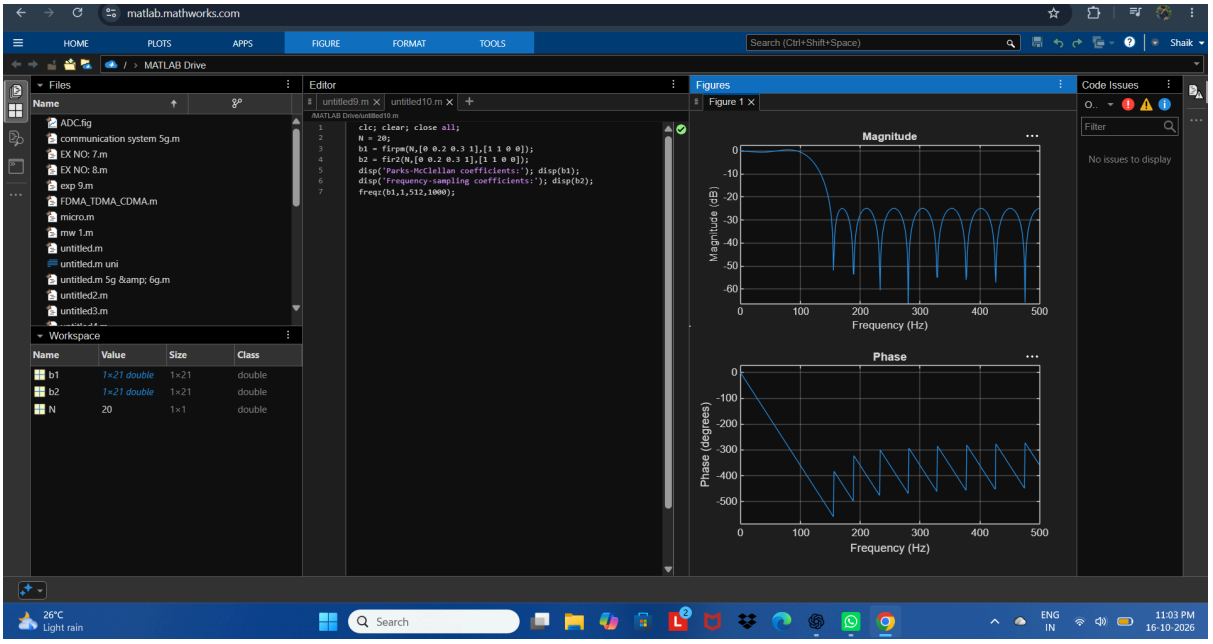
```
clc; clear; close all;
b = fir1(20,0.2,hamming(21));
x = [1 zeros(1,49)];
y1 = filter(b,1,x);
[sos,g] = tf2sos(b,1);
y2 = g*x;
for k = 1:size(sos,1)
    y2 = filter(sos(k,1:3),sos(k,4:6),y2);
end
plot(y1); hold on; plot(y2,'--');
legend('Direct','Cascade'); grid on;
```

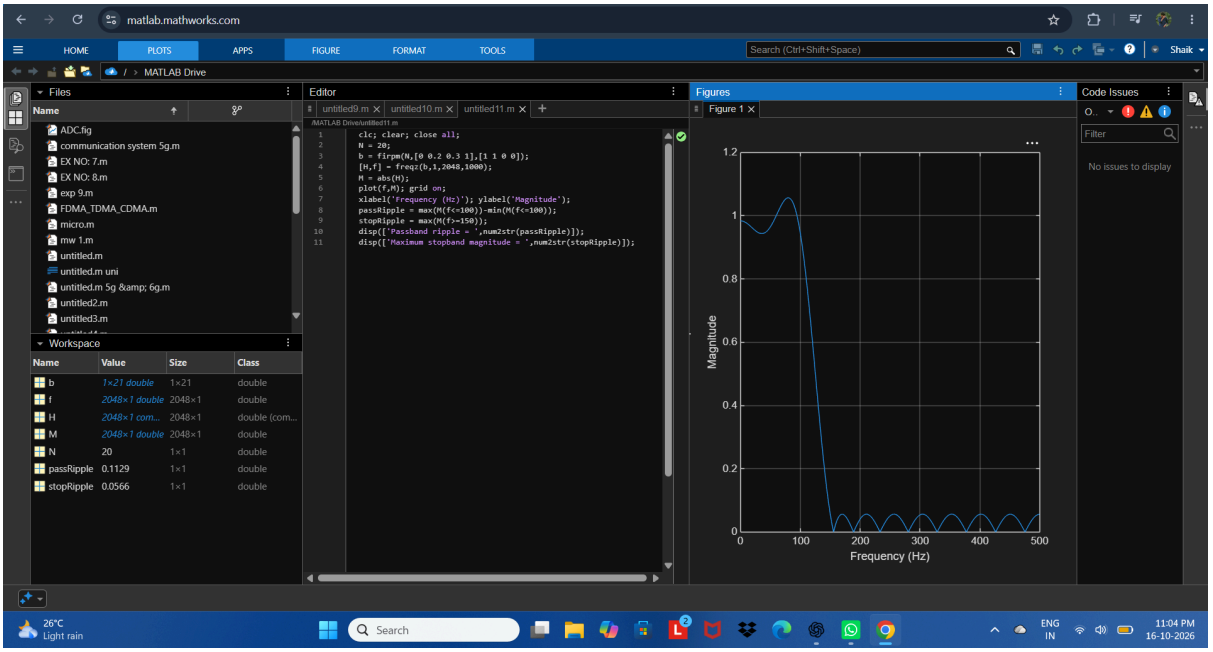
```
clc; clear;
b = fir1(20,0.2,hamming(21));
disp('FIR coefficients:'); disp(b);
disp(['Symmetric: ',num2str(isequal(b,fliplr(b)))]);
```

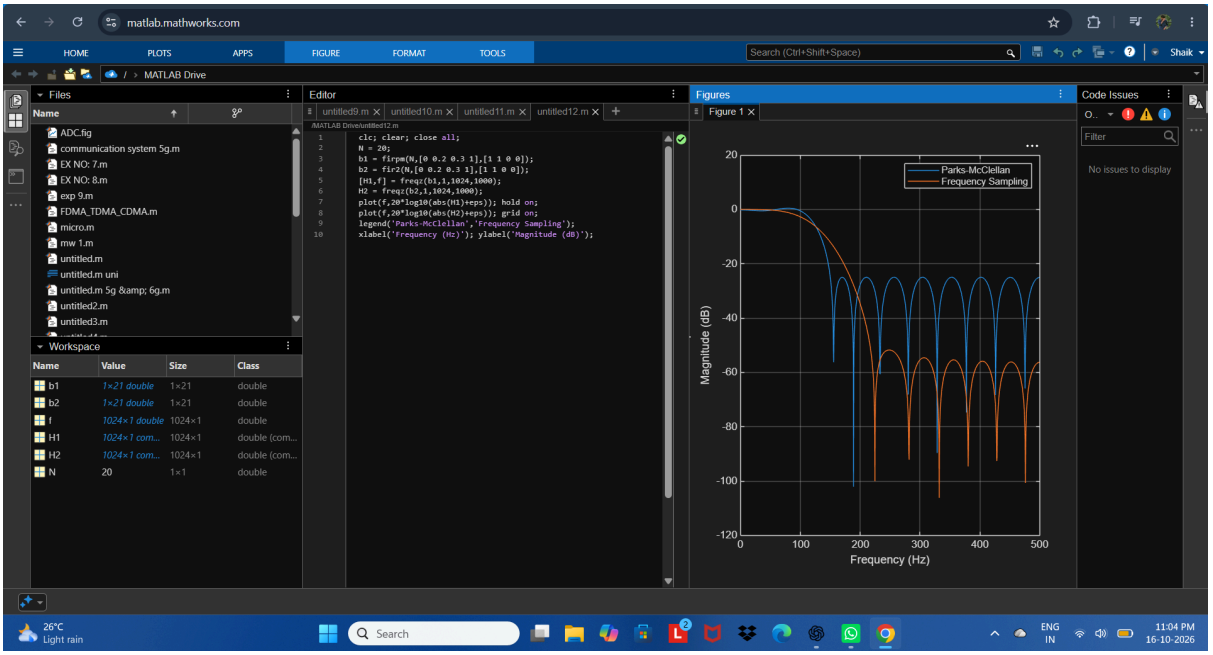
```
clc; clear; close all;
b = fir1(20,0.2,hamming(21));
x = [1 zeros(1,99)];
y = filter(b,1,x);
[H,f] = freqz(b,1,1024,1000);
subplot(3,1,1); stem(y); title('Impulse Response');
subplot(3,1,2); plot(f,unwrap(angle(H))); grid on;
```

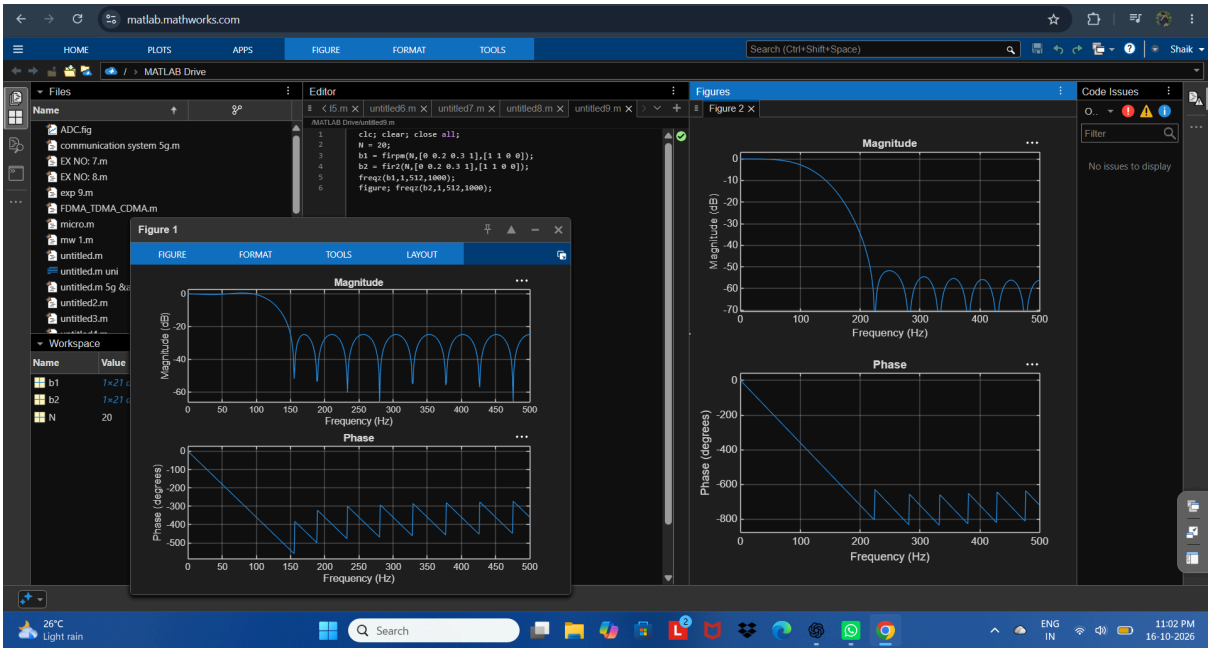
```
title('Phase Response');
subplot(3,1,3); plot(y); grid on;
title('Output Response');

clc; clear; close all;
b = fir1(20,0.2,hamming(21));
x = randn(1,100);
y1 = filter(b,1,x);
[sos,g] = tf2sos(b,1);
y2 = g*x;
for k = 1:size(sos,1)
    y2 = filter(sos(k,1:3),sos(k,4:6),y2);
end
disp(['Maximum difference = ',num2str(max(abs(y1-y2)))]);
plot(y1); hold on; plot(y2,'--');
legend('Direct','Cascade'); grid on;
```


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EX NO: 8.m
exp 9.m
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micro.m
mw 1.m
untitled.m
untitled.m uni
untitled.m 5g & 6g.m
untitled2.m
untitled3.m
Workspace
Name
Value
Size
Class
b1
1×21 double
1×21
double
b2
1×21 double
1×21
double
N
20
1×1
double
Editor
untitled9.m
untitled10.m
clc; clear; close all;
N = 20;
b1 = firpm(N,[0 0.2 0.3 1],[1 1 0 0]);
b2 = firl2(N,[0 0.2 0.3 1],[1 1 0 0]);
disp('Parks-McClellan coefficients:'); disp(b1);
disp('Frequency-sampling coefficients:'); disp(b2);
freqz(b1,1,512,1000);
Figures
Figure 1
Magnitude
Magnitude (dB)
Frequency (Hz)
Phase
Phase (degrees)
Frequency (Hz)
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untitled.m uni
untitled.m 5g & 6g.m
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untitled3.m
Workspace
Name
Value
Size
Class
b
1×21 double
1×21
double
f
2048×1 double
2048×1
double
H
2048×1 com...
2048×1
double (com...
M
2048×1 double
2048×1
double
N
20
1×1
double
passRipple
0.1129
1×1
double
stopRipple
0.0566
1×1
double
Editor
untitled9.m
untitled10.m
untitled11.m
clc; clear; close all;
N = 20;
b = firpm(N,[0 0.2 0.3 1],[1 1 0 0]);
[H,f] = freqz(b,1,2048,1000);
M = abs(H);
plot(f,M); grid on;
xlabel('Frequency (Hz)'); ylabel('Magnitude');
passRipple = max(M(f<=100))-min(M(f<=100));
stopRipple = max(M(f>=150));
disp(['Passband ripple = ',num2str(passRipple)]);
disp(['Maximum stopband magnitude = ',num2str(stopRipple)]);
Figures
Figure 1
Magnitude
Frequency (Hz)
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Parks-McClellan
Frequency Sampling
Magnitude (dB)
Frequency (Hz)


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Phase (degrees)
Frequency (Hz)